

Course Code : ECT451-2

UVXV/MW – 24 / 1663

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

MICROWAVE THEORY AND TECHNIQUES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated.
 - (3) Illustrate your answers with neat sketches.
 - (4) Assume suitable data wherever necessary.
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1. (a) Enlist applications of various Microwave frequency band in different fields according to its property. 5(CO1)
 - (b) Justify, why vacuum tubes are not used in Microwave frequency bands. 5(CO5)
 2. (a) Discuss single stub Impedance matching techniques used in transmission lines. Write its merits and demerits. 5(CO2)
 - (b) Suggest the IEEE frequency band over which WR90 waveguide can be used near to dominant mode TE_{10} . 5(CO2)
 3. (a) Interpret working principle of Electronic Mixer. 5(CO3)
 - (b) Design a Rectangular Microstrip Antenna (RMSA) at 2.3 GHz. Choose a substrate with dielectric constant of 2.2 loss tangent of 0.02 and thickness of 1.6 mm. 5(CO3)
 4. (a) Describe the effect of repeller voltage on modes of operation of Reflex Klystron. Draw necessary waveforms. 5(CO4)

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- (b) Prove that H-Plane tee work as a 3 dB power splitter. 5(CO4)
5. (a) Illustrate High VSWR measurement Technique in detail. 5(CO5)
- (b) Two identical directional couplers of 30 dB are used to sample incident and reflected power in a waveguide. VSWR = 2 and the output of the coupler sampling incident power = 4.5 mW. Determine the reflected power ? 5(CO5)
6. (a) Draw and discuss the block diagram of RADAR system. 5(CO1)
- (b) Determine the maximum range of radar system which operates at 3 cm with peak pulse power of 600 KW. If its antenna is 5 m², minimum detectable signal power is 10⁻¹³ W and the radar cross sectional area of the target is 20 m². 5(CO1)

